Lie Algebras

The Exceptional Series of Lie Groups II

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Abstract. Numerology of exceptional Lie groups and their representations.

As in [1] and [2], we consider in this Note the groups G of the exceptional series

$$\mathrm{e},\ A_1,\ A_2,\ G_2,\ D_4,\ F_4,\ E_6,\ E_7,\ E_8.$$

Between the trivial group and A_1 we also insert the supergroup $\mathrm{SOSp}(1,2)$. More than the group itself, what interests us each time is its category of representations. For $\mathrm{SOSp}(1,2)$, we consider the category of super-representations for which the parity is given by $-1 \in \mathrm{Sp}(2) \subset \mathrm{SOSp}(1,2)$. The neutral component G^0 of each group is the adjoint group, and in each case the category of representations under consideration is tensor-generated by the adjoint representation.

The dual Coxeter number $h^\vee$ of G^0 is, according to type,

$$6/5,\ 3/2,\ 2,\ 3,\ 4,\ 6,\ 9,\ 12,\ 18,\ 30.$$

For the trivial group, $h^\vee = 6/5$ by fiat. We regard G as depending on the parameter $h^\vee$. Sometimes it is more convenient to use $\mu := 6/h^\vee$, which we now prefer to the parameter λ used in [1] and [2]:

$$\mu = 5,\ 4,\ 3,\ 2,\ 3/2,\ 1,\ 2/3,\ 1/2,\ 1/3,\ 1/5.$$

For G^0 of rank r, consider the following dominant weights $\theta_1, \dots, \theta_r$. For E_8 we stop at θ_7. Notation: $(a_1, \dots, a_r)$ denotes $\sum a_i\theta_i$, the fundamental weights ω_i being taken in the order of the Bourbaki tables. For G^0 of rank at least 1, that is, $G \neq \mathrm{e}$, θ_1 is the highest root, corresponding to the adjoint representation. This defines the θ_i for G of rank at most 1, namely $G^0 = \mathrm{e}$, $\mathrm{SOSp}(1,2)$, or A_1. After that, we take

A_2: $(1,1)$, $(0,3)$;

G_2: $(0,1)$, $(3,0)$;

D_4: $(0,1,0,0)$, $(1,0,1,1)$, $(2,0,0,0)$, $(2,0,2,0)$;

F_4: $(1,0,0,0)$, $(0,1,0,0)$, $(0,0,0,2)$, $(0,0,2,0)$;

E_6: $(0,1,0,0,0,0)$, $(0,0,0,1,0,0)$, $(1,0,0,0,0,1)$, $(0,0,1,0,1,0)$, $(0,0,0,0,1,1)$, $(1,0,0,0,2,0)$;

E_7: the fundamental weights ω_1, ω_3, ω_6, ω_4, and $(0,1,0,0,0,0,1)$, $(0,1,0,0,1,0,0)$, $(0,0,0,0,1,0,1)$;

E_8: the fundamental weights ω_8, ω_7, ω_1, ω_6, ω_2, ω_5, ω_3.

For A_2, D_4, and E_6, it would be immaterial to replace these θ_i by their images under a common automorphism of the Dynkin diagram.

When G is connected, we write $\mathcal{R}(p_1, \ldots, p_r)$ for the irreducible representation of G of dominant weight $\sum p_i\theta_i$. When G is disconnected, this weight must be decorated, as in [1], in order to define $\mathcal{R}(p_1, \ldots, p_r)$. The required decoration is as follows. For A_1: the sign $(-1)^{p_1}$ if $p_2 = 0$, and 0 otherwise. For D_4: $+$, $0+$, or $0-$. For E_8: $+$ or $0-$.

Let V_i be the representation $\mathcal{R}(p_1, \ldots, p_r)$ for $p_i = \delta_i^j$. The representation V_1, defined for G of rank at least 1, is the adjoint representation. The representations V_i ($1 \le i \le 7$) are the representations $\mathcal{J}$, $\mathcal{X}_2$, $\mathcal{Y}_2^*$, $\mathcal{X}_3$, $\mathcal{O}^*$, $\mathcal{X}_4$, $\mathcal{F}^*$ of [1]. The representation $\mathcal{R}(p_1, \ldots, p_r)$ is characterized by the properties that it has dominant weight $\sum p_i\theta_i$ and is a constituent of the tensor product of the $V_i^{\otimes p_i}$.

For $s \le r$, we will also write $\mathcal{R}(p_1, \ldots, p_s)$ for $\mathcal{R}(p_1, \ldots, p_s, 0, \ldots, 0)$.

Our purpose is to describe, for fixed $p_1, \ldots, p_7$, similarities among the representations $\mathcal{R}(p_1, \ldots, p_s)$ of the groups in the exceptional series of rank at least s. The analogue in the series A is this: given a decreasing sequence of integers $\ell_1 \ge \ell_2 \ge \cdots \ge \ell_k \ge 0$, take the corresponding representation of $\mathrm{GL}(n)$ for each $n \ge k$.

Let G be a simple split pinned group, let $\mathfrak{g}$ be its Lie algebra, and let $\mathfrak{h}$ be the Lie algebra of the pinned split torus T. Identify the group of weights $X := \mathrm{Hom}(T, \mathbb{G}_\mathrm{m})$ with a subgroup of the dual $\mathfrak{h}^\vee$. The canonical bilinear form $(\ ,\ )$ is the bilinear form on $\mathfrak{h}^\vee$ inverse to the Killing form $\mathrm{Tr}(\mathrm{ad}\, x \cdot \mathrm{ad}\, y)$ on $\mathfrak{h}$. For a classical supergroup, for instance $\mathrm{SOSp}(1, 2)$, use the supertrace. If α_0 is the highest root, one definition of the dual Coxeter number $h^\vee$ is that

$$(\alpha_0, \alpha_0) = 1/h^\vee.$$

To each dominant weight λ we will attach a finite system of rational numbers, each taken with a multiplicity $m(q) \in \mathbb{Z}$. In other words, we attach a measure $L_G(\lambda)$ on $\mathbb{Q}$, a $\mathbb{Z}$-linear combination $\sum m(q)\delta[q]$ of Dirac measures. Let $[R^+]$ be the measure on X given by the sum of the Dirac measures $\delta[\alpha]$ for α a positive root. Let $M(\lambda)$ be the measure on $\mathbb{Q}$ which is the image of $[R^+]$ under $(2\lambda, \cdot) : X \to \mathbb{Q}$. The factor "12" is explained by the integrality assertion in Theorem 1. If all roots have the same length, or if α is a long root, then $(\lambda, \alpha) = \lambda(H_\alpha)/2h^\vee$. If G has roots of two different lengths and α is short, then

$$(\lambda, \alpha) = \big((\alpha, \alpha)/(\alpha_0, \alpha_0)\big)\lambda(H_\alpha)/2h^\vee.$$

The factor $(\alpha, \alpha)/(\alpha_0, \alpha_0)$ is 1/4, 1/3, 1/2 for $\mathrm{SOSp}(1, 2)$, G_2, and F_4, respectively. Let ρ be the half-sum $\frac{1}{2}\int \alpha \cdot [R^+]$ of the positive roots. For a classical supergroup, for example $\mathrm{SOSp}(1, 2)$, these definitions should use instead the sum over positive roots $\sum m(\alpha)\delta[\alpha]$, with $m(\alpha) = 1$ if the corresponding root subspace is even and $m(\alpha) = -1$ if it is odd. We define

$$L_G(\lambda) := M(\lambda + \rho) - M(\rho). \tag{1}$$

When no ambiguity can result, we simply write $L(\lambda)$ for $L_G(\lambda)$. If $L(\lambda) = \sum m(q)\delta[q]$, Weyl's dimension formula says that the representation $V(\lambda)$ of dominant weight λ has dimension

$$\dim V(\lambda) = \prod_q m(q)^{-1}. \tag{2}$$

Let $F_{2N}^{\mathfrak{g}}$ be the function $\mathrm{Tr}((\mathrm{ad}\, x)^{2N})$ on $\mathfrak{g}$. Restrict it to $\mathfrak{h}$, and identify $\mathfrak{h}$ with $\mathfrak{h}^\vee$ by the canonical bilinear form. We obtain a function $F_{2N}^{\mathfrak{g}}$ on $\mathfrak{h}^\vee$. By construction,

$$F_{2N}^{\mathfrak{g}}(\lambda + \rho) - F_{2N}^{\mathfrak{g}}(\rho) = 2 \cdot 12^{-N} \sum m(q) q^{2N} L_G(\lambda). \tag{3}$$

The factor 2 comes from the fact that the definition of $L_G(\lambda)$ starts from R^+ and not from the set $R = R^+ \cup (-R^+)$ of all roots. Applying Theorem 2, p. 267, of Duflo [3] to the weights λ and

0, one can make explicit an element of the center of the enveloping algebra which acts on $V(\lambda)$ by $F^{\mathfrak{g}}_{2N}(\lambda+\rho) - F^{\mathfrak{g}}_{2N}(\rho)$. For $N=1$, it is the Casimir.

For $p_1,\ldots,p_7 \in \mathbb{N}$, we will construct a formal linear combination $L(p_1,\ldots,p_7)$ of elements of the space $\mathcal{L}$ of linear forms $x \mapsto Ax+B$ with integral coefficients. In other words, it is a $\mathbb{Z}$-linear combination $\sum m(A,B)\delta[Ax+B]$ of Dirac measures on $\mathbb{Q}$. This combination will specialize to the measure $L_G(\sum p_i\theta_i)$ attached to G.

For $N=1$, that is, for the scalar by which the Casimir acts on $\mathcal{R}(p_1,\ldots,p_7)$, this is even a linear function:

Theorem 1. *With the preceding notation, for arbitrary $p_1,\ldots,p_7$, the sum $\sum m(A,B)(Ax+B)^2$ is linear in x; equivalently,*

$$\sum m(A,B)A^2 = 0. \tag{4}$$

For $r \le 7$, we write $L(p_1,\ldots,p_r)$ for $L(p_1,\ldots,p_r,0,\ldots,0)$ with $7-r$ final zeroes. Fix $r \le 7$ and a group G in the exceptional series of rank at least r. To G there corresponds a rational number μ.

Theorem 2. *If $p_1,\ldots,p_r \ge 0$, the zero linear form is not in the support of $L(p_1,\ldots,p_r)$, and the total mass of the linear forms $Ax+B$ such that $A\mu+B=0$ is zero. The scalar*

$$\sum_{A\mu+B=0} (A\mu+B)m(A,B)$$

is an integer. It is the length of the restriction to G^0 of $\mathcal{R}(p_1,\ldots,p_r)$. According to (2), $\dim \mathcal{R}(p_1,\ldots,p_r)$ *is therefore the value at μ of the rational function $\prod(Ax+B)^{-m(A,B)}$.*

More precisely, if $r \le \operatorname{rank}(G)$ and $A\mu+B=0$, the multiplicity $m(A,B)$ of $Ax+B$ in $L(p_1,\ldots,p_r)$ is zero except in the following cases. If $G=A_2$ and $p_2>0$: 1 for $-2x+4$ and -1 for $-2x+2$. If $G=D_4, E_8$ and either $p_3>0$, $p_4=0$, or $p_3=0$, $p_4>0$: 1 for $-3x+3$ and -1 for $-x+1$. If $G=F_4, E_8$ and $p_3,p_4>0$: 1 for $-3x+3$ and $-2x+2$, and -2 for $-x+1$. If $G=E_6$ and p_5 or p_6 is >0: 1 for $-4x+2$ and -1 for $-2x+1$.

The $\mathcal{R}(p_1,\ldots,p_7)$ for

$$p_1+2p_2+2p_3+3p_4+3p_5+3p_6+4p_7 \le 4$$

are the representations 1, $\mathcal{J}$, $\mathcal{A}$, $\mathcal{C}$, $\mathcal{O}^*$, $\mathcal{D}$, $\mathcal{E}$, $\mathcal{F}$, $\mathcal{F}^*$, $\mathcal{G}$, $\mathcal{H}$, $\mathcal{I}$, $\mathcal{J}$, $\mathcal{X}_i$ $(i\le 4)$, $\mathcal{Y}_i$ $(i \le 4)$, and $\mathcal{Y}_i^*$ of [1], which may be read off from E_7 or E_8. Thus Theorem 3 recovers part of the dimension formulae of [1].

Let $\mathcal{V}(p_1,\ldots,p_7)$ be the convolution

$$\mathcal{V}(p_1,\ldots,p_7) := L(p_1,\ldots,p_7) * (\delta[x]-\delta[x-1]). \tag{5}$$

If $L(p_1,\ldots,p_7) = \sum m(A,B)\delta[Ax+B]$ and $\mathcal{V}(p_1,\ldots,p_7) = \sum n(A,B)\delta[Ax+B]$, then the finitely supported functions $m(A,B)$ and $n(A,B)$ determine one another by

$$n(A,B) = m(A,B) - m(A+1,B), \tag{6}$$

$$m(A,B) = \sum_{A' \ge A} n(A',B). \tag{7}$$

In (10), the sum is over A'; the factor $(A' \ge A)$ is 1 if $A' \ge A$ and 0 otherwise.

We will take (10) as the definition of $L(p_1,\ldots,p_7)$, with $\mathcal{V}(p_1,\ldots,p_7)$ defined by specialization from a $\mathbb{Z}$-linear combination $\mathbf{V}$ of Dirac measures on the space of maps from $\mathbb{Z}^7$ to $\mathbb{Q}$ of the form

$$(B;n;a_1,\ldots,a_7) \longmapsto \left(2\sum a_ip_i + n\right)x + B. \tag{8}$$

The measure $\mathbf{V}$ has a simple description in terms of the root system of type E_8.

Write $(B; n; a_1, \ldots, a_7)$ for the function (11), write $\delta[B; n; a_1, \ldots, a_7]$ for the Dirac measure supported at this function, and write $\{\varepsilon_1, \ldots, \varepsilon_8\}$ for the root $2\delta - \alpha_3$ of E_8. With this notation, $\mathbf{V}$ is defined as follows.

(A) Each positive root $\{c, e, g, h, f, d, b, a\}$ fixed by s_4 contributes to $\mathbf{V}$

$$\delta[B + h + n - 1; a, b, c, d, e, f, g] - \delta[B + h; n - 1; a, b, c, d, e, f, g]$$

with $5h + n = a + b + c + d + e + f + g + h$.

(B) Each pair of positive roots $\{c, e, g, h', f, d, b, a\}$, $\{c, e, g, h'', f, d, b, a\}$ interchanged by s_4, with $h'' < h'$—then $h'' = h' - 1$—contributes to $\mathbf{V}$

$$\delta[B + h' + n' - 1; a, b, c, d, e, f, g] - \delta[B + h''; n'' - 1; a, b, c, d, e, f, g]$$

with $5h' + n' = a + b + c + d + e + f + g + h'$ and $5h'' + n'' = a + b + c + d + e + f + g + h''$.

(C) $\mathbf{V}$ is the sum of these contributions and of the $\mathbb{Z}$-linear combination of the $\delta[B; n; 0, \ldots, 0]$ such that, for every B and n, the total mass of the $(B; n; a_i, \ldots)$ is zero.

Definition 3. (i) $\mathcal{V}(p_1, \ldots, p_7)$ is the image of $\mathbf{V}$ under evaluation at $(p_1, \ldots, p_7)$:

$$(B; n; a_1, \ldots, a_7) \longmapsto \left(2 \sum a_i p_i + n\right) x + B \in \mathbb{Q}.$$

(ii) It follows from (C) above that, for each B_0, the mass in $\mathcal{V}(p_1, \ldots, p_7)$ of the $Ax + B$ with $B = B_0$ is zero. Define $L(p_1, \ldots, p_7)$ by (8), i.e. by (10).

It is immediate from the definition of $\mathbf{V}$ that

Proposition 4. (i) *On the support of* $\mathbf{V}$, *one has* $B \in \mathbb{Z}$ *and* $n \geq 0$.

(ii) *The projections of* $\mathbf{V}$ *by*

(a) $(B; n; a_1, \ldots, a_7) \mapsto (B; n) \in \mathbb{Z}^2$,

(b) $(B; n; a_1, \ldots, a_7) \mapsto (a_1, \ldots, a_7) \in \mathbb{Z}^7$,

are zero.

The vanishing (ii a) is equivalent to $\mathcal{V}(0, \ldots, 0) = 0$, i.e. to $L(0, \ldots, 0) = 0$. This is compatible with the fact that $\mathcal{R}(0, \ldots, 0)$ is, for every G, the trivial representation of dominant weight 0, and that $L_G(0) = 0$.

Taking account of (1) and of the fact that the total mass of $\mathbf{V}$ is zero, the vanishing (ii b) is equivalent to the following statement: for every N, as a function of $p_1, \ldots, p_7$,

$$\sum \left(2 \sum a_i p_i + B\right)^{2N} L(p_1, \ldots, p_7)$$

has degree at most $2N$. This is compatible, through (4), with the fact that for every simple Lie group G, $F^{\mathfrak{g}}_{2N}(\lambda)$ is a function of degree at most $2N$ in the weight λ.

Put $\Lambda = 2 \sum a_i \theta_i + \rho$. Theorem 2 can be reformulated as

$$\int_\Lambda (\Lambda + x - 1)(\Lambda + 1) \cdot \mathbf{V} = 0, \tag{9}$$

identically in the indeterminates p_i.

Some indications on the proof of Theorem 1. — Let G be a group of the exceptional series and put $r = \inf(\operatorname{rank}(G), 7)$. Except for E_8 and the simple root α_4, no positive root is orthogonal to all the θ_i $(1 \le i \le r)$. Put $[R^G] := [R^+]$ if $G \neq E_8$, and $[R^G] = [R^+] - \delta[\alpha_4]$ if $G = E_8$. Define the moving part of $L_G(\sum p_i\theta_i)$ by

$$L_G^{\mathrm{mob}}\left(\sum p_i\theta_i\right) = \left\langle M\left(\sum p_i\theta_i + \rho\right), [R^G]\right\rangle. \tag{10}$$

The definition (1) rewrites as $L_G(\sum p_i\theta_i) = L_G^{\mathrm{mob}}(\sum p_i\theta_i) - L_G^{\mathrm{mob}}(0)$.

Theorem 1 for G uses of $\mathbf{V}$ only its image $\mathbf{V}_G$ under

$$(B; n; a_1, \ldots, a_7) \mapsto \left(2\sum a_i p_i + n + B; a_1, \ldots, a_r\right). \tag{11}$$

Decompose $\mathbf{V}_G$ as $\mathbf{V}_G^+ - \mathbf{V}_G^-$, the moving part $\mathbf{V}_G^+$, respectively the fixed part $\mathbf{V}_G^-$, having support where $(a_1, \ldots, a_r) \neq 0$, respectively $= 0$. Theorem 1 follows from its analogue for the moving parts of L and $\mathbf{V}$: by Proposition 1 (ii a), the fixed part is recovered by putting $x = 0$. For $(a_1, \ldots, a_r) \neq 0$, let $R^{(a_1,\ldots,a_r)}$ be the set of positive roots α such that $(\alpha, \theta_i) = a_i$ for $1 \le i \le r$. The form $(\ ,\ )$ is as in (5)–(6). For $G \neq E_8$, each $R^{(a_1,\ldots,a_r)}$ is a singleton or empty, and their union is R^+. For $G = E_8$, each $R^{(a_1,\ldots,a_r)}$ is an orbit of $\{s_3, s_4\}$ or is empty. Their union is $R^+ - \{\alpha_4\}$. Let $\mathbf{V}_G^{(a_1,\ldots,a_r)}$ be the measure on $\mathbb{Q}$ which is the inverse image of $\mathbf{V}_G$ by $q \mapsto (Bh^\vee/6 + n; a_1, \ldots, a_r)$: the mass of q is the mass of the $(B; n; a_1, \ldots, a_r, \ldots)$ with $Bh^\vee/6 + n = q$. Let $[R^{(a_1,\ldots,a_r)}]$ be the sum of the $\delta[\alpha]$ for $\alpha \in R^{(a_1,\ldots,a_r)}$. Theorem 1, for the moving parts of L and $\mathbf{V}$, decomposes as the sum of the equalities

$$(p_i \cdot [R^{(a_1,\ldots,a_r)}]) * (\delta[0] - \delta[-1]) = \mathbf{V}_G^{(a_1,\ldots,a_r)}. \tag{12}$$

Our proof of (15) is by brute force. Below are a few properties of the root systems involved which may help explain what is happening.

When all roots have the same length, $(\lambda, \alpha) = \lambda(H_\alpha)$ and (ρ, α) is the sum of the coefficients of α expressed as a linear combination of simple roots. For E_8, $(\rho, \alpha) = 5$, and (15) follows immediately from the definition of $\mathbf{V}$.

Let R be the root system of type A_1, A_2, D_4, E_6, or E_7, and let r be its rank. In the root system of type E_8, the fundamental weights $\theta_1, \ldots, \theta_r$ correspond to simple roots $\alpha_{S(1)}, \ldots, \alpha_{S(r)}$. Let S_1 be the complementary set of $8 - r$ simple roots, and let W' be the Weyl group generated by the corresponding reflections. Let $\langle\theta_1, \ldots, \theta_r\rangle$ be the vector space spanned by $\theta_1, \ldots, \theta_r$. Other descriptions: it is the orthogonal of $\langle\theta_i \mid i \in S_1\rangle$, or the fixed subspace of W'. One checks case by case that the roots of E_8 lying in $\langle\theta_1, \ldots, \theta_r\rangle$ form a root system of type R, and that, if one takes as positive roots of this root system those that are positive for E_8, the fundamental weights of E_8 are the dominant weights $\theta_1, \ldots, \theta_r$ of R.

For R of one of the indicated types, (15) follows from the following assertions, which we verified for D_4, F_4, and E_7:

(a) Let T be an orbit of W' in the set of roots of E_8. If T is not reduced to a fixed point and, for $t \in T$, (t, θ_i)—constant on T—is > 0 for some θ_i $(1 \le i \le r)$, then the contribution of T to $\mathbf{V}$ has zero image under (14).

(b) Let ρ_R and $h_R^\vee$ denote the sum of the fundamental weights and the dual Coxeter number of R. If α lies in $\langle\theta_1, \ldots, \theta_r\rangle$ and is a positive root, necessarily fixed by s_4, then, with the notation of part (A) of the definition of $\mathbf{V}$, one has

$$(h^\vee/6)(\rho_R, \alpha) + n = (\rho, \alpha).$$

Examples. — Verify (a) for E_7. The orbit T is here a pair of roots as in part (B) of the definition of $\mathbf{V}$. The point is to verify that $3h' + n' \ge 3h'' + n'' - 1$. Indeed, $h'' = h' - 1$ and $4h' + n' = 4h'' + n''$, i.e. $n'' = n' + 4$.

Theorem 1 gives the dependence on G of $F_{2N}^{\mathfrak{g}}(\lambda + \rho) - F_{2N}^{\mathfrak{g}}(\rho)$ for certain λ.

Proposition 5. *For $1 \leq N \leq 8$, $F_{2N}(\rho)$ is, as a function of G, a function of $\mu = 6/h^\vee$ of the form $P_{2N}(\mu)/\mu(\mu+1)$, for a polynomial $P_{2N}(X)$ of degree $2N$ satisfying $P_{2N}(X) = P_{2N}(-1-X)$.*

It follows from Duflo [3] that $F_{2N}(\rho)$ is the value for G, in the sense of [5], of a linear combination of trivalent graphs with oriented vertices. For $N \leq 3$, arguments imitating [5], 6.8, 6.9, make it possible to deduce the proposition from this presentation of $F_{2N}(\rho)$. Beyond that, we know how to verify it only by brute force. Since we have only ten groups G at our disposal, the proposition remains true, but is empty, for $N \geq 9$.

Correction to [2]. — The formula given for $\dim(\mathcal{X}_4)$ is erroneous. See [1] for the correct formula.

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